

NORTH SYDNEY GIRLS HIGH SCHOOL



Mathematics Extension 2 2013 Trial HSC Examination

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen.
Black pen is preferred
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11-16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section 1 – Pages 2 – 5

10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – Pages 6-14

90 marks

- Attempt Questions 11 - 16
- Allow about 2 hours and 45 minutes for this section

Student Number _____

Class _____

Student Name _____

QUESTION	MARK
1 – 10	/10
11	/15
12	/15
13	/15
14	/15
15	/15
16	/15
TOTAL	/100

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

1 Let $z = a + ib$ where a and b are real and non-zero. Which of the following is not true?

- (A) $z + \bar{z}$ is real
- (B) $\frac{z}{\bar{z}}$ is non-real
- (C) $z^2 - (\bar{z})^2$ is real
- (D) $z\bar{z}$ is real and positive

2 Which of the following corresponds to the set of points in the complex plane defined by $|z + 2i| = |z|$?

- (A) the point given by $z = -i$
- (B) the line $\text{Im}(z) = -1$
- (C) the circle with centre $-2i$ and radius 1
- (D) the line $\text{Re}(z) = -1$

3 The equation $9x^3 - 27x^2 + 11x - 7 = 0$ has roots α , β and γ .

What is the value of $\frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma} + \frac{1}{\alpha\beta}$?

- (A) $\frac{27}{7}$
- (B) $-\frac{27}{7}$
- (C) $-\frac{11}{7}$
- (D) $\frac{11}{7}$

4 The polynomial equation $P(x) = 0$ has real coefficients, and has roots which include $x = -2 + i$ and $x = 2$. What is the minimum possible degree of $P(x)$?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

- 5 Using a suitable substitution, what is the correct expression for $\int_0^{\frac{\pi}{3}} \sin^3 x \cos^4 x \, dx$ in terms of u ?

(A) $\int_0^{\frac{\sqrt{3}}{2}} (u^4 - u^6) \, du$

(B) $\int_1^{\frac{1}{2}} (u^6 - u^4) \, du$

(C) $\int_{\frac{1}{2}}^1 (u^6 - u^4) \, du$

(D) $\int_0^{\frac{\sqrt{3}}{2}} (u^6 - u^4) \, du$

- 6 There are 5 pairs of socks in a drawer. Four socks are randomly chosen from the drawer. Which expression represents the probability that all four of the socks come from different pairs ?

(A) $1 \times \frac{1}{9} \times \frac{1}{8} \times \frac{1}{7}$

(B) $1 \times \frac{9}{10} \times \frac{8}{9} \times \frac{7}{8}$

(C) $1 \times \frac{8}{9} \times \frac{6}{7} \times \frac{4}{5}$

(D) $1 \times \frac{8}{9} \times \frac{6}{8} \times \frac{4}{7}$

- 7 The equation $x^3 + y^3 = 3xy$ is differentiated implicitly with respect to x . Which of the following expressions is $\frac{dy}{dx}$?

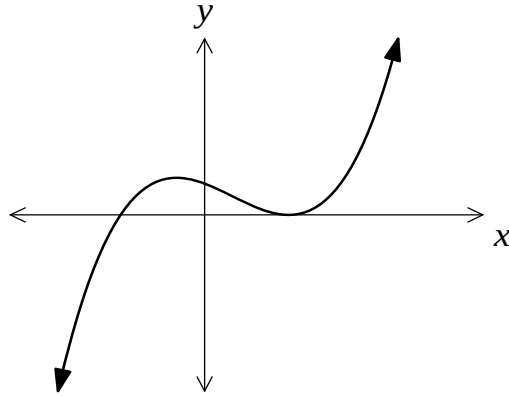
(A) $\frac{y - x^2}{y^2 - x}$

(B) $\frac{y^2 - x}{y - x^2}$

(C) $\frac{x^2 + y^2}{x}$

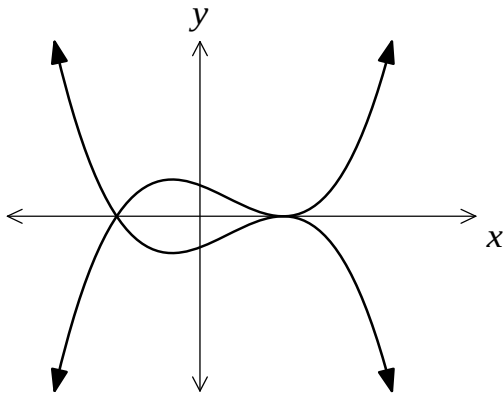
(D) $\frac{x^2}{x - y^2}$

- 8 The graph $y = f(x)$ is shown.

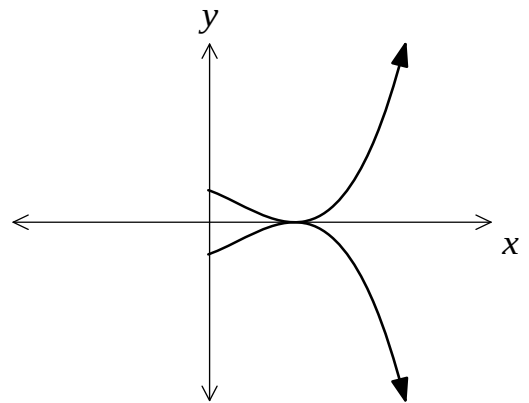


Which of the following graphs best represents $y^2 = f(x)$?

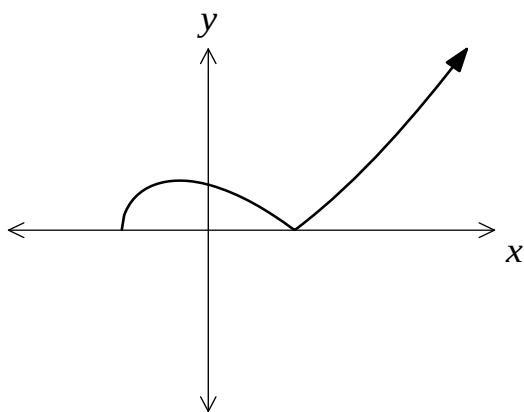
(A)



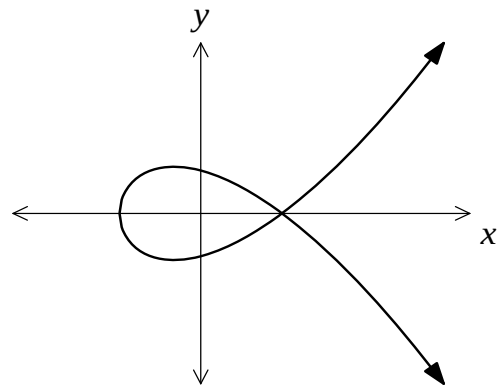
(B)



(C)



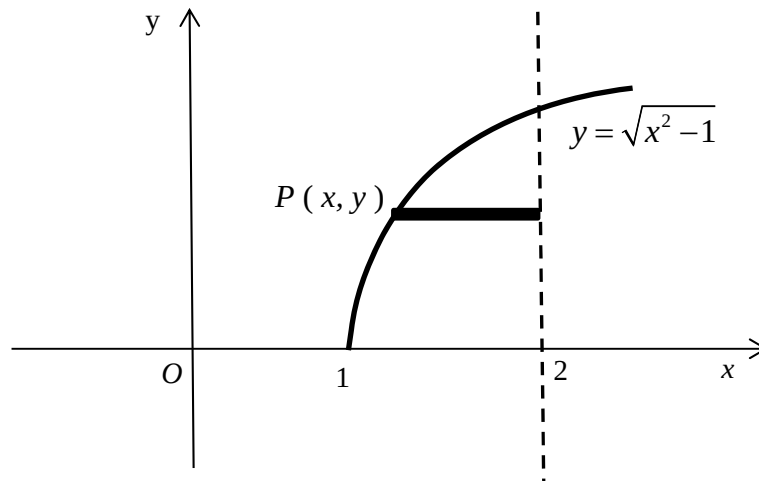
(D)



- 9 A body is moving in a straight line and, after t seconds, it is x metres from the origin and travelling at $v \text{ ms}^{-1}$. Given that $v = x$ and that $t = 3$ where $x = -1$, what is the equation for x in terms of t ?

- (A) $x = e^{t-3}$
 (B) $x = -e^{t-3}$
 (C) $x = \sqrt{2t-5}$
 (D) $x = -\sqrt{2t-5}$

10



The region bounded by the x axis, the curve $y = \sqrt{x^2 - 1}$ and the line $x = 2$ is rotated around the y axis.

The slice at $P(x, y)$ on the curve is perpendicular to the axis of rotation. What is the volume δV of the annular slice formed?

- (A) $\pi(3 - y^2)\delta y$
 (B) $\pi(4 - (y^2 + 1)^2)\delta y$
 (C) $\pi(4 - x^2)\delta x$
 (D) $\pi(2 - x)^2\delta x$

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Let $z = -5 - 12i$ and $w = 2 - i$. Find in the form $x + iy$

(i) $(1+i)\bar{w}$ 1

(ii) $\frac{z}{2-3i}$ 2

(b) By first writing $w = -\sqrt{3} + i$ in modulus argument form, show that $w^3 - 8i = 0$. 2

(c) By completing the square, find $\int \frac{1}{\sqrt{3-2x-x^2}} dx$. 2

(d) Use the substitution $u = x^2 + 1$ to evaluate $\int_0^{\sqrt{3}} \frac{x^3}{\sqrt{x^2+1}} dx$. 3

(e) (i) Without using calculus, sketch the curve $y = \frac{x+2}{(x-1)(x+3)}$ showing all important features. 2

(ii) Find the area bounded by the curve and the x -axis between $x = 2$ and $x = 5$. 3

Question 12 (15 marks) Use a SEPARATE writing booklet.

- a) Consider the complex number $z = x + iy$ where $z^2 = a + ib$.
- (i) Sketch on the same set of axes, the graphs of $x^2 - y^2 = a$ and $2xy = b$ where both a and b are positive. 2
The foci and directrices of the curves need NOT be found.
- (ii) Use the graphs to explain why there are two distinct square roots of the complex number $a + ib$ if $a > 0$ and $b > 0$. 1
- (iii) Consider how the sketch changes when b is negative. What is the relationship between the new square roots and those found when b was positive? 1
- (b) The region enclosed by the curves $y = \frac{4}{x^2 + 4}$ and $y = \frac{1}{x^2 + 1}$ and the ordinates $x = 0$ and $x = 2$ is rotated about the y axis. Using the method of cylindrical shells, find the volume of the solid formed. 3
- (c) A particle's acceleration is given by $\ddot{x} = 3(1 - x)(1 + x)$ where x is the displacement in metres. Initially the particle is at the origin with velocity 2 metres per second.
- (i) Show that $v^2 = 2(2 - x)(x + 1)^2$. 2
- (ii) Find the velocity and acceleration at $x = 2$. 2
- (iii) Describe the motion of the particle. 2
- (iv) Find the maximum speed and where it occurs. 2

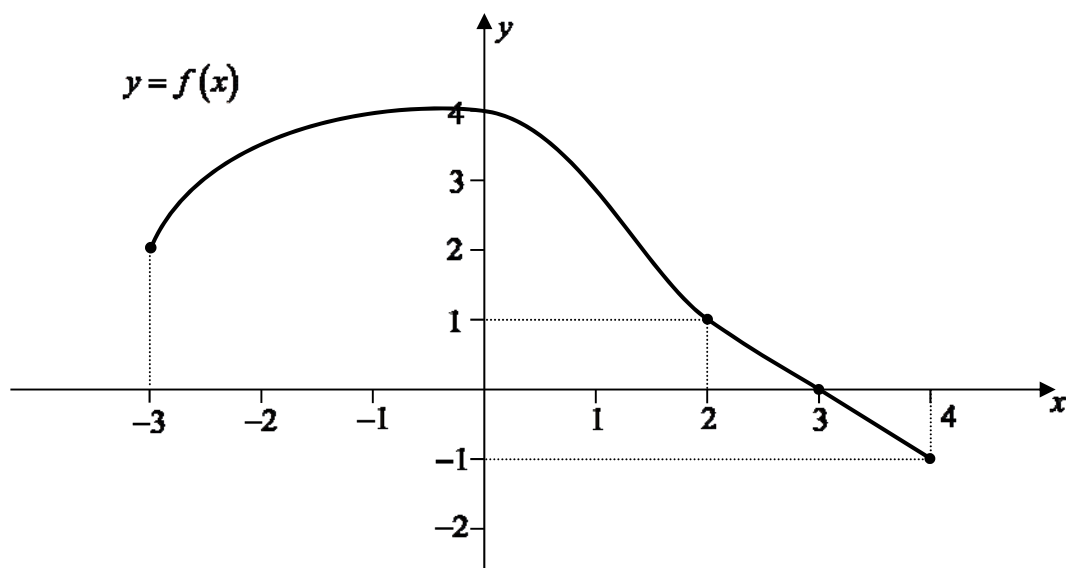
Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Show that $x^2 + y^2 \geq xy$ where x and y are real numbers. 2
- (ii) If $x + y = 3z$ show that $x^2 + y^2 \geq 3z^2$. 2
- (b) The complex numbers z and w each have a modulus of 2. The arguments of z and w are $\frac{4\pi}{9}$ and $\frac{7\pi}{9}$ respectively.
- (i) Sketch vectors representing z , w and $z + w$ on the Argand diagram, showing any geometrical relationships between the three vectors. 2
- (ii) Find $\arg(z + w)$. 1
- (iii) Evaluate $|z + w|$. 1
- (c) (i) Use the substitution $t = \tan \frac{x}{2}$ to show that $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \sin x} = \frac{\pi}{3\sqrt{3}}$. 3
- (ii) Show that $\int_0^{2a} f(x) dx = \int_0^a [f(x) + f(2a - x)] dx$. 2
- (iii) Hence, or otherwise, evaluate $\int_0^{\pi} \frac{x}{2 + \sin x} dx$. 2

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Question 14 (15 marks) Use a SEPARATE writing booklet.

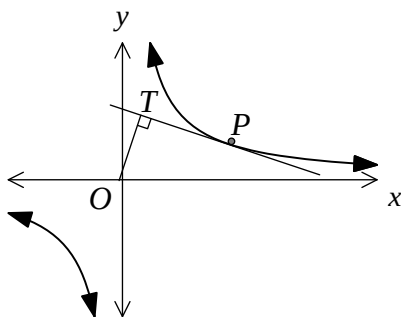
- (a) The diagram shows the graph of $y = f(x)$ which is only defined over the domain $-3 \leq x \leq 4$.



Draw separate one-third page sketches of the graphs of the following:

- | | | |
|------|-----------------|---|
| (i) | $y = f(x)$ | 1 |
| (ii) | $y = \ln(f(x))$ | 2 |

(b)



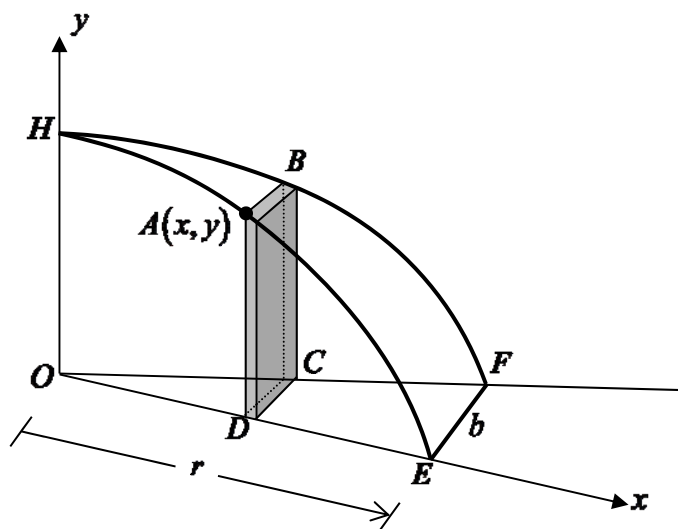
The point $P\left(ct, \frac{c}{t}\right)$ lies on the hyperbola $xy = c^2$. The point T lies at the foot of the perpendicular drawn from the origin O to the tangent at P .

- | | | |
|-------|---|---|
| (i) | Show that the tangent at P has equation $x + t^2y = 2ct$. | 2 |
| (ii) | If the coordinates of T are (x_1, y_1) show that $y_1 = t^2x_1$. | 1 |
| (iii) | Show that the locus of T is given by $(x^2 + y^2)^2 = 4c^2xy$. | 2 |

Question 14 continues on page 11

Question 14 (continued)

(c)



The horizontal base of a solid is an isosceles triangle $OE F$ where $OE = OF = r$ and $EF = b$. HAE is the parabolic arc with equation $y = r^2 - x^2$ where E lies on the x -axis. HBF is another parabolic arc, congruent to HAE , so that the plane $O H B F$ is vertical. A rectangular slice $A B C D$ of width δx is taken perpendicular to the base, such that $C D$ lies in the base and $C D \parallel E F$.

(i) Show that the volume of the slice $A B C D$ is $\frac{bx}{r}(r^2 - x^2)\delta x$. 2

(ii) Hence show that the solid $HOEF$ has volume $\frac{br^3}{4}$. 2

(iii) Suppose now that $\angle EOF = \frac{2\pi}{n}$ and that n identical solids $HOEF$ are arranged about O as centre with common vertical axis OH to form a solid S . Show that 2

the volume V_n of S is given by $V_n = \frac{1}{2}r^4n \sin \frac{\pi}{n}$.

(iv) When n is large, the solid S approximates the volume of the solid of revolution formed by rotating the region bound by the x axis and the curve $y = r^2 - x^2$ about the y axis. 1

Using the fact that $\frac{\sin x}{x} \rightarrow 1$ as $x \rightarrow 0$ find $\lim_{n \rightarrow \infty} V_n$.

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) The equation $2x^3 - 5x + 1 = 0$ has roots α, β, γ . Find the equation whose roots are $-2\alpha, -2\beta$, and -2γ . 2

- (b) (i) For $z = \cos \theta + i \sin \theta$, show that $z^n + z^{-n} = 2 \cos n\theta$. 2

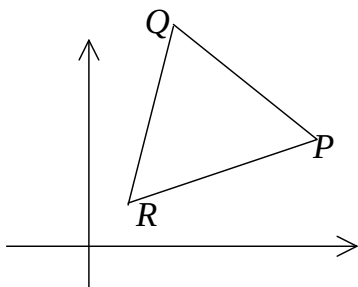
- (ii) If $z + \frac{1}{z} = u$, find an expression for $z^3 + \frac{1}{z^3}$ in terms of u . 2

- (iii) It can be shown that $z^5 + \frac{1}{z^5} = u^5 - 5u^3 + 5u$. (Do not prove this). 3

Show that

$$1 + \cos 10\theta = 2(16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta)^2$$

- (c)



In the Argand diagram, the points P, Q and R represent the complex numbers p, q and r .

- (i) Given that the triangle PQR is equilateral, explain why 1

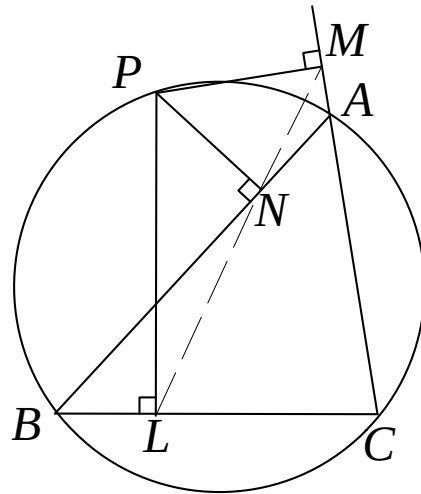
$$r - q = \operatorname{cis} \frac{2\pi}{3} (q - p)$$

- (ii) Hence, or otherwise show $2r = (p + q) + i\sqrt{3}(q - p)$ 1

Question 15 continues on page 13

Question 15 (continued)

(d)



In the diagram, P is any point on the circle ABC . The point N lies on AB such that PN is perpendicular to AB . Similarly, points M and L lie at the foot of the perpendiculars drawn from P to CA (produced) and BC respectively.

- | | | |
|------|--|----------|
| (i) | State why $BLNP$ is a cyclic quadrilateral. | 1 |
| (ii) | Prove that the points L , M and N are collinear. | 3 |

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) Let $P(a \cos \theta, b \sin \theta)$ be a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The focus of the ellipse is $S(ae, 0)$ where e is the eccentricity and O is the origin.
- (i) Find the coordinates of the centre C and the radius of the circle of which SP is a diameter. 2
- (ii) Show that $OC = \frac{a}{2}(e \cos \theta + 1)$ 2
- (b) (i) Show that the polynomial $P(x) = 4x^3 + 10x^2 + 8x + 3$ is divisible by $(2x + 3)$. 1
- (ii) Hence express the polynomial in the form $P(x) = A(x)Q(x)$ where $Q(x)$ is a real quadratic polynomial. 2
- (c) Using the fact that $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy}$ for $0 \leq x < 1$ and $0 \leq y < 1$ 3
 prove by mathematical induction that for all positive integers n ,

$$\tan^{-1} \frac{1}{2 \times 1^2} + \tan^{-1} \frac{1}{2 \times 2^2} + \tan^{-1} \frac{1}{2 \times 3^2} + \dots + \tan^{-1} \frac{1}{2 \times n^2} = \frac{\pi}{4} - \tan^{-1} \frac{1}{2n+1}$$

Question 16 continues on page 15

Question 16 (continued)

(d) Consider $f(x) = \log x - x + 1$.

(i) Show that $f(x) \leq 0$ for all $x > 0$. **2**

(ii) Consider the set of n positive numbers $p_1, p_2, p_3, \dots, p_n$ such that **1**

$$p_1 + p_2 + p_3 + \dots + p_n = 1.$$

By using the result in part (i), deduce that

$$\sum_{r=1}^n \log(np_r) \leq np_1 + np_2 + np_3 + \dots + np_n - n$$

(iii) Show that $\sum_{r=1}^n \log np_r \leq 0$. **1**

(iv) Hence deduce that $0 < n^n p_1 p_2 p_3 \dots p_n \leq 1$ **1**

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

2013 Extension 2 Trial HSC Solutions

1. (A) $z + \bar{z} = (a + ib) + (a - ib)$

$$= 2a \quad (\text{which is real})$$

(But you should know that $z + \bar{z} = 2 \operatorname{Re} z$)

(B) $\frac{z}{\bar{z}} = \frac{a + ib}{a - ib} \times \frac{a + ib}{a + ib}$

$$= \frac{a^2 - b^2}{a^2 + b^2} + \frac{2ab}{a^2 + b^2}i \quad (\text{which is not real since } a, b \neq 0)$$

Alternatively

$$\frac{z}{\bar{z}} = \frac{r \operatorname{cis} \theta}{r \operatorname{cis}(-\theta)}$$

$$= \operatorname{cis} 2\theta$$

which is real if $2\theta = k\pi$ (where k is an integer)

$$\theta = \frac{k}{2}\pi \quad (\text{ie. } z \text{ is either real or pure imaginary})$$

But this is not the case, since neither a nor b is zero

(C) $z^2 - (\bar{z})^2 = (a + ib)^2 - (a - ib)^2$

$$= a^2 + 2abi - b^2 - a^2 + 2abi + b^2$$

$$= 4abi \quad (\text{which is never real since } a, b \neq 0)$$

(D) $z\bar{z} = |z|^2$ which by definition of modulus is real and positive

(C)

2. Without doing any algebra, this is the set of point which are equidistant from $(0, -2)$ and $(0, 0)$.

ie. $y = -1$

(B)

3. $\frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma} + \frac{1}{\alpha\beta} = \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma}$

(A)

$$= \frac{\frac{27}{9}}{\frac{7}{9}}$$

$$= \frac{27}{7}$$

4. Since $P(x) = 0$ has real coefficients, the conjugate of the root $x = -2 + i$ must also be a root.

(C)

So the polynomial must have at least 3 roots (and there is not enough information to conclude more.)

5. $\int_0^{\frac{\pi}{3}} \sin^3 x \cos^4 x dx = \int_0^{\frac{\pi}{3}} \cos^4 x (1 - \cos^2 x) \cdot \sin x dx$

Let $u = \cos x$

$x = 0, u = 1$

(B)

$du = -\sin x dx$

$x = \frac{\pi}{3}, u = \frac{1}{2}$

$$= \int_1^{\frac{1}{2}} u^4 (1 - u^2) \cdot (-du)$$

$$= \int_1^{\frac{1}{2}} (u^6 - u^4) du$$

6. 1st sock can be any of them.
 2nd sock cannot be the only matching sock – 8 possibilities of 9 socks remaining
 3rd sock cannot be either of the two matching socks – 6 possibilities of the 8 socks remaining
 4th sock cannot be either of the three matching socks – 4 possibilities of the 7 socks remaining **(D)**

7. $x^3 + y^3 = 3xy$

$$\cancel{x}^2 + \cancel{y}^2 \cdot \frac{dy}{dx} = \cancel{x} \cdot \frac{dy}{dx} + y \cdot \cancel{y} \quad (\text{product rule})$$

$$(y^2 - x) \frac{dy}{dx} = y - x^2$$

$$\frac{dy}{dx} = \frac{y - x^2}{y^2 - x}$$

Alternative setting out:

$$x^3 + y^3 = 3xy \quad \textbf{(A)}$$

$$\cancel{x}^2 \cdot dx + \cancel{y}^2 \cdot dy = \cancel{x} (x \cdot dy + y \cdot dx)$$

$$(y^2 - x) dy = (y - x^2) dx$$

$$\frac{dy}{dx} = \frac{y - x^2}{y^2 - x}$$

8. Negative root (single root) must become a vertical point.
 To the left of the negative root, $f(x)$ is -ve, so can't be square rooted.
 Positive root is a multiple root, so we can't determine the nature of the corresponding point on the new graph. (But we are only asked for the *best* answer.)
 $y^2 = f(x)$ becomes $y = \pm \sqrt{f(x)}$, hence symmetry in the x -axis. **(D)**
9. Easiest method – check the options by differentiating to get v .
 The only options whose derivatives are the function itself (ie. $v = x$) are (A) and (B).
 But (B) is the only option that also allows x to equal -1 .

Alternative method:

$$\frac{dx}{dt} = x$$

$$\frac{dx}{x} = dt$$

$$\int \frac{dx}{x} = \int dt$$

$$\ln|x| = t + c$$

$$(t = 3, x = -1) \quad 0 = 3 + c$$

$$c = -3$$

$$\ln|x| = t - 3$$

$$|x| = e^{t-3}$$

$$x = \pm e^{t-3}$$

But for x to equal -1 , we need the -ve case:

$$x = -e^{t-3} \quad \textbf{(B)}$$

$$\begin{aligned}
 10. \quad \delta V &= \pi(R^2 - r^2)h \\
 &= \pi(2^2 - x^2)\delta y
 \end{aligned}$$

$$\begin{aligned}
 \text{But } y &= \sqrt{x^2 - 1} \\
 y^2 &= x^2 - 1 \\
 x^2 &= y^2 + 1
 \end{aligned}$$

$$\begin{aligned}
 \therefore \delta V &= \pi[4 - (y^2 + 1)]\delta y \\
 &= \pi(3 - y^2)\delta y
 \end{aligned} \tag{A}$$

Question 11

$$\begin{aligned}
 (a) \quad (i) \quad (1+i)\overline{w} &= (1+i)(2+i) \\
 &= 2 + i + 2i - 1 \\
 &= 1 + 3i
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \frac{z}{2-3i} &= \frac{-5-12i}{2-3i} \times \frac{2+3i}{2+3i} \\
 &= \frac{-10-15i-24i+36}{4+9} \\
 &= \frac{26-39i}{13} \\
 &= 2-3i
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad w &= 2 \operatorname{cis} \frac{5\pi}{6} \\
 w^3 - 8i &= \left(2 \operatorname{cis} \frac{5\pi}{6}\right)^3 - 8i \\
 &= 8 \operatorname{cis} \frac{5\pi}{2} - 8i \\
 &= 8i - 8i \quad \left(\text{hopefully you don't need to write } \operatorname{cis} \frac{5\pi}{2} \text{ as } \cos \frac{5\pi}{2} + i \sin \frac{5\pi}{2} \text{ to see this}\right) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad \int \frac{dx}{\sqrt{3-2x-x^2}} &= \int \frac{dx}{\sqrt{-(x^2+2x+1)+3+1}} \\
 &= \int \frac{dx}{\sqrt{4-(x+1)^2}} \\
 &= \sin^{-1} \frac{x+1}{2} + c
 \end{aligned}$$

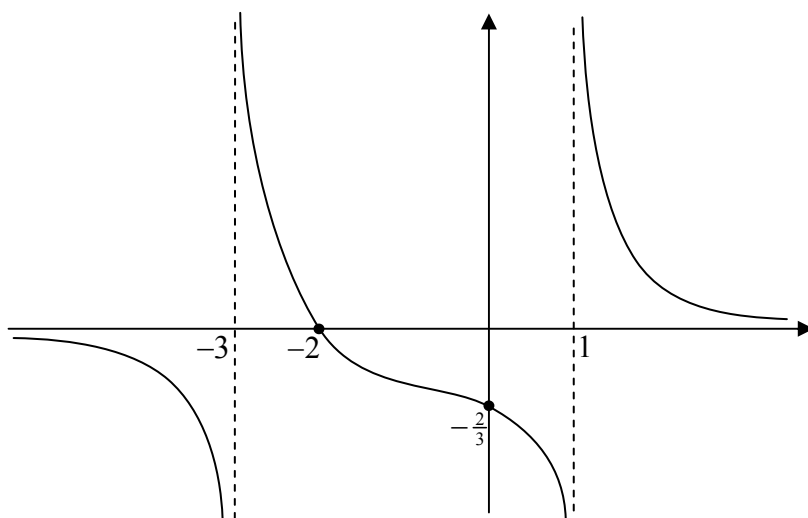
$$\begin{aligned}
 \text{(d)} \quad \int_0^{\sqrt{3}} \frac{x^3 dx}{\sqrt{x^2+1}} &= \frac{1}{2} \int_0^{\sqrt{3}} \frac{x^2}{\sqrt{x^2+1}} \cdot 2x dx \\
 &= \frac{1}{2} \int_1^4 \frac{u-1}{\sqrt{u}} \cdot du \\
 &= \frac{1}{2} \int_1^4 \left(u^{\frac{1}{2}} - u^{-\frac{1}{2}} \right) du \\
 &= \left[\frac{1}{3} u^{\frac{3}{2}} - u^{\frac{1}{2}} \right]_1^4 \\
 &= \frac{8}{3} - 2 - \frac{1}{3} + 1 \\
 &= \frac{4}{3}
 \end{aligned}$$

$$\text{Let } u = x^2 + 1$$

$$du = 2x dx$$

$$\begin{pmatrix} x=0, & u=1 \\ x=\sqrt{3}, & u=4 \end{pmatrix}$$

(e) (i)



$$\text{(ii)} \quad A = \int_2^5 \frac{(x+2)dx}{(x-1)(x+3)}$$

$$\text{Let } \frac{x+2}{(x-1)(x+3)} = \frac{A}{x-1} + \frac{B}{x+3}$$

$$x+2 = A(x+3) + B(x-1)$$

$$(x=1) \quad 3 = 4A \Rightarrow A = \frac{3}{4}$$

$$(x=-3) \quad -1 = -4B \Rightarrow B = \frac{1}{4}$$

$$A = \frac{1}{4} \int_2^5 \left(\frac{3}{x-1} + \frac{1}{x+3} \right) dx$$

$$= \frac{1}{4} \left[3 \ln|x-1| + \ln|x+3| \right]_2^5$$

$$= \frac{1}{4} (3 \ln 4 + \ln 8 - 0 - \ln 8)$$

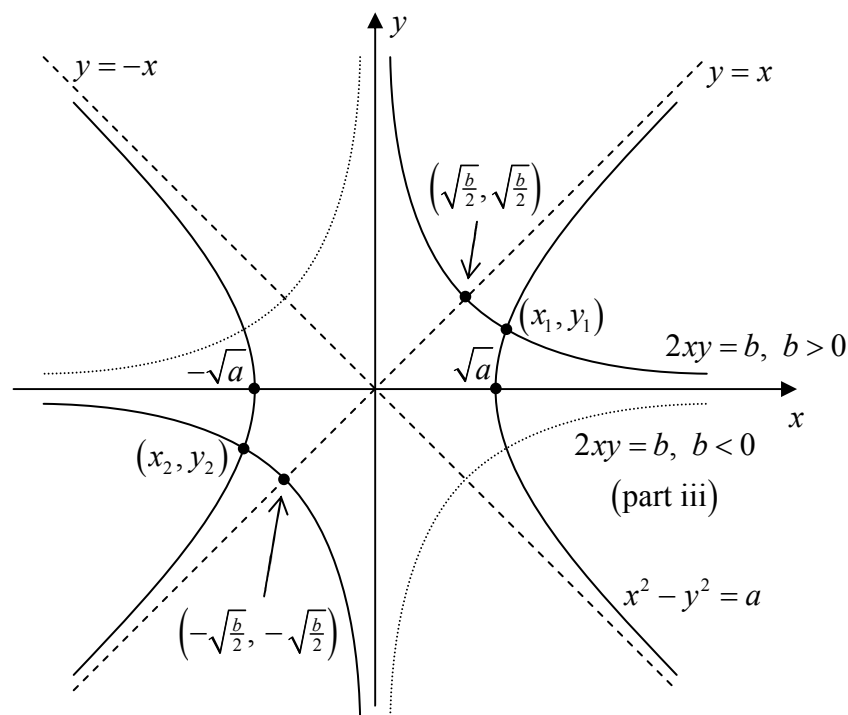
$$= \frac{1}{4} \ln \frac{4^3 \times 8}{5}$$

$$= \frac{1}{4} \ln \frac{512}{5}$$

Question 12

- (a) (i) When $y = x$, $2x^2 = b$ ($b > 0$)

$$x = \pm \sqrt{\frac{b}{2}}$$



- (ii) Let $z = x + iy$

$$z^2 = a + ib$$

$$(x + iy)^2 = a + ib$$

$$(x^2 - y^2) + 2xyi = a + ib$$

Equating real and imaginary parts: $x^2 - y^2 = a$

$$2xy = b$$

Solving simultaneously for x and y , we get the graphs of part (i).

The graphs show that there are two distinct points of intersection (x_1, y_1) and (x_2, y_2)

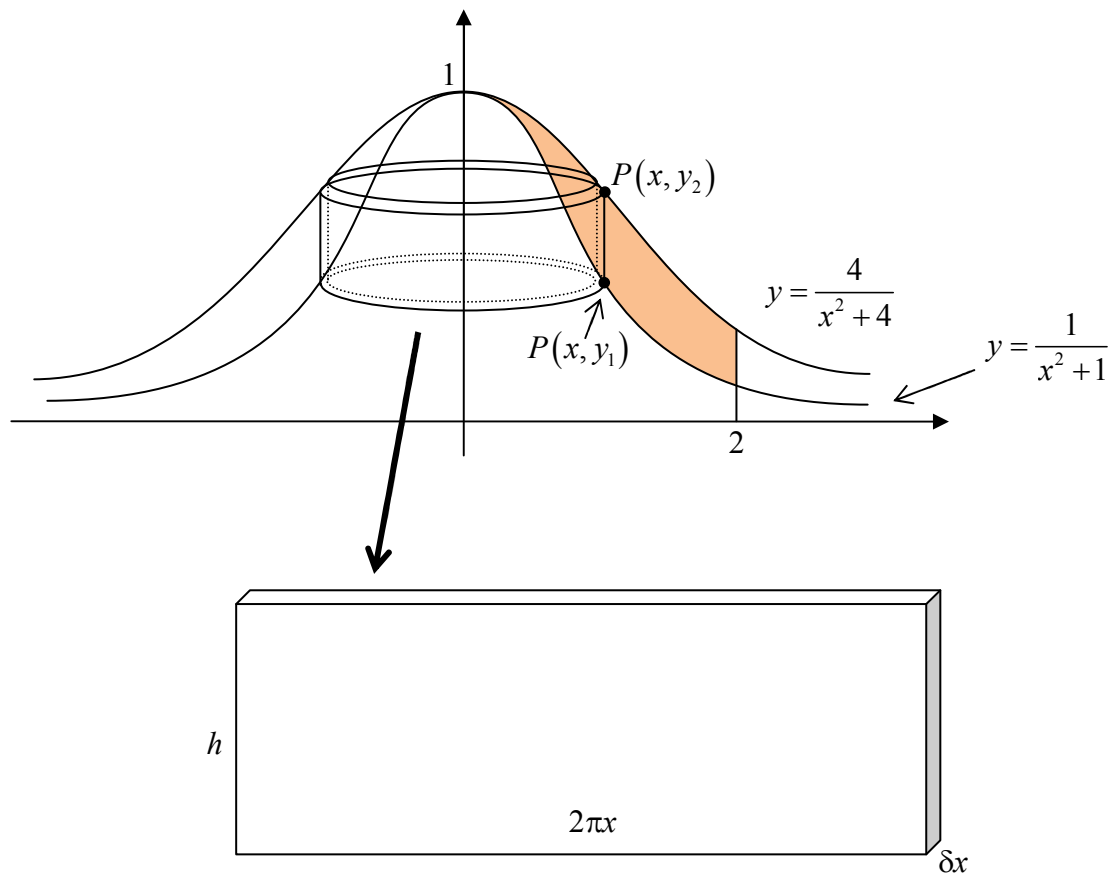
corresponding to two distinct complex roots $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ of the complex number $a + ib$.

- (iii) When b is negative, the graph of $2xy = b$ lies in the 2nd and 4th quadrants.

So the points of intersection with $x^2 - y^2 = a$ are $(x_1, -y_1)$ and $(x_2, -y_2)$.

ie. the new square roots are the conjugates of the roots found in part (ii).

(b)



$$h = y_2 - y_1$$

$$= \frac{4}{x^2 + 4} - \frac{1}{x^2 + 1}$$

Volume of shell $\delta V \approx 2\pi x h \cdot \delta x$

$$= 2\pi x \left(\frac{4}{x^2 + 4} - \frac{1}{x^2 + 1} \right) \delta x$$

$$\text{Volume } V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^2 2\pi x \left(\frac{4}{x^2 + 4} - \frac{1}{x^2 + 1} \right) \delta x$$

$$= \pi \int_0^2 \left(\frac{8x}{x^2 + 4} - \frac{2x}{x^2 + 1} \right) dx$$

$$= \pi \left[4 \ln(x^2 + 4) - \ln(x^2 + 1) \right]_0^2$$

$$= \pi (4 \ln 8 - \ln 5 - 4 \ln 4 + 0)$$

$$= \pi \ln \frac{8^4}{5 \times 4^4}$$

$$= \pi \ln \frac{16}{5} \text{ units}^3$$

(c) (i) $\ddot{x} = 3(1-x)(1+x)$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 3 - 3x^2$$

$$\frac{1}{2} v^2 = 3x - x^3 + c$$

$$(x=0, v=2) \quad 2 = c$$

$$\frac{1}{2} v^2 = 3x - x^3 + 2$$

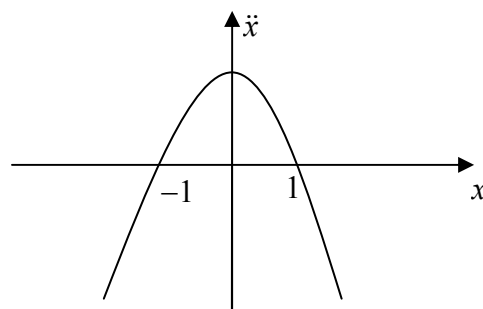
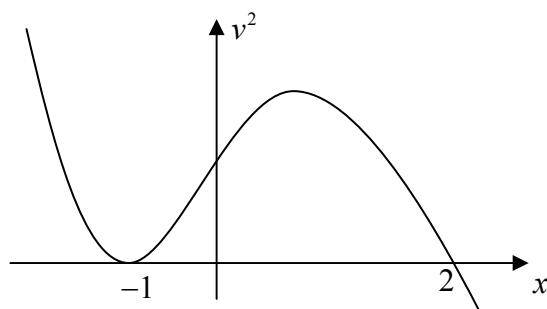
$$v^2 = 6x - 2x^3 + 4$$

Check RHS: $2(2-x)(x+1)^2 = (4-2x)(x^2+2x+1)$
 $= 4x^2 + 8x + 4 - 2x^3 - 4x^2 - 2x$
 $= 6x - 2x^3 + 4$
 $= \text{LHS}$

$$\therefore v^2 = 2(2-x)(x+1)^2$$

(ii) $x = 2: \quad v = 0$
 $\ddot{x} = 3(1-2)(1+2)$
 $= -9 \text{ ms}^{-2}$

(iii)



Firstly, the particle cannot ever be to the right of $x = 2$, as v^2 would be $-ve$.

Secondly, the particle can *possibly* change direction only when $v = 0$, ie. at $x = -1$ and $x = 2$.

Initially, the velocity is $+2$, so the particle moves to the right, speeding up until it reaches $x = 1$, then slowing to a stop at $x = 2$.

Since the acceleration at $x = 2$ is $-ve$, it then changes direction, speeds up until it again reaches $x = 1$, then slowing to a stop at $x = -1$.

At $x = -1$ the velocity and acceleration are both zero (and dependent only on position, not time), so the particle remains at $x = -1$.

(iv) From the graphs, the max speed (over the restricted domain $-1 \leq x \leq 2$) occurs at $x = 1$ ($\ddot{x} = 0$).

$$v_{\max}^2 = 2(2-1)(1+1)^2$$

$$= 8$$

$$v_{\max} = 2\sqrt{2} \text{ m/s}$$

Question 13

(a) (i) $(x - y)^2 \geq 0 \quad \forall \text{ real } x, y$
 $x^2 - 2xy + y^2 \geq 0$
 $x^2 + y^2 \geq 2xy$

If x, y have the same sign, $xy > 0$, so $2xy > xy$, so $x^2 + y^2 \geq xy$.

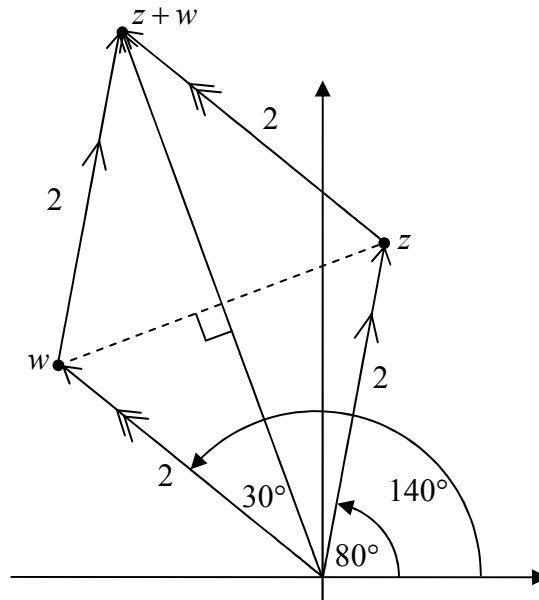
If x, y have opposite sign, $xy < 0$, so $x^2 + y^2 \geq xy$ as $x^2 + y^2 \geq 0$.

OR

$$\begin{aligned} x^2 + y^2 - xy &= \left(x - \frac{y}{2}\right)^2 + \frac{3}{4}y^2 \\ &\geq 0 \quad (\text{sum of 2 perfect squares}) \\ x^2 + y^2 &\geq xy \end{aligned}$$

(ii)
$$\begin{aligned} x^2 + y^2 - 3z^2 &= x^2 + y^2 - 3\left(\frac{x+y}{3}\right)^2 \\ &= x^2 + y^2 - \frac{x^2 + 2xy + y^2}{3} \\ &= \frac{3x^2 + 3y^2 - x^2 - 2xy - y^2}{3} \\ &= \frac{2x^2 + 2y^2 - 2xy}{3} \\ &= \frac{2}{3}(x^2 + y^2 - xy) \\ &\geq 0 \quad (\text{since } x^2 + y^2 \geq xy \text{ from i}) \\ x^2 + y^2 &\geq 3z^2 \end{aligned}$$

(b) (i)



(ii) Since this shape is a rhombus, the vertex angles are bisected by the diagonals.

$\therefore \arg(z+w)$ is the average of $\frac{4\pi}{9}$ and $\frac{7\pi}{9}$

$$\begin{aligned} \arg(z+w) &= \frac{1}{2} \left(\frac{4\pi}{9} + \frac{7\pi}{9} \right) & \left[\text{OR } \arg(z+w) = \frac{4\pi}{9} + \frac{1}{2} \left(\frac{7\pi}{9} - \frac{4\pi}{9} \right) \right] \\ &= \frac{11\pi}{18} \end{aligned}$$

(iii) Since diagonals bisect each other at right angles, $\frac{1}{2}|z+w| = 2 \cos 30^\circ$
 $|z+w| = 2\sqrt{3}$

(c) (i)
$$\begin{aligned} \int_0^{\frac{\pi}{2}} \frac{dx}{2+\sin x} &= \int_0^1 \frac{\frac{2dt}{1+t^2}}{2+\frac{2t}{1+t^2}} \times \frac{1+t^2}{1+t^2} \\ &= \int_0^1 \frac{2dt}{2(1+t^2)+2t} \\ &= \int_0^1 \frac{dt}{t^2+t+1} \\ &= \int_0^1 \frac{dt}{\left(t+\frac{1}{2}\right)^2 + \frac{3}{4}} \\ &= \frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{2\left(t+\frac{1}{2}\right)}{\sqrt{3}} \right]_0^1 \\ &= \frac{2}{\sqrt{3}} \left(\tan^{-1} \sqrt{3} - \tan^{-1} \frac{1}{\sqrt{3}} \right) \\ &= \frac{2}{\sqrt{3}} \left(\frac{\pi}{3} - \frac{\pi}{6} \right) \\ &= \frac{\pi}{3\sqrt{3}} \end{aligned}$$

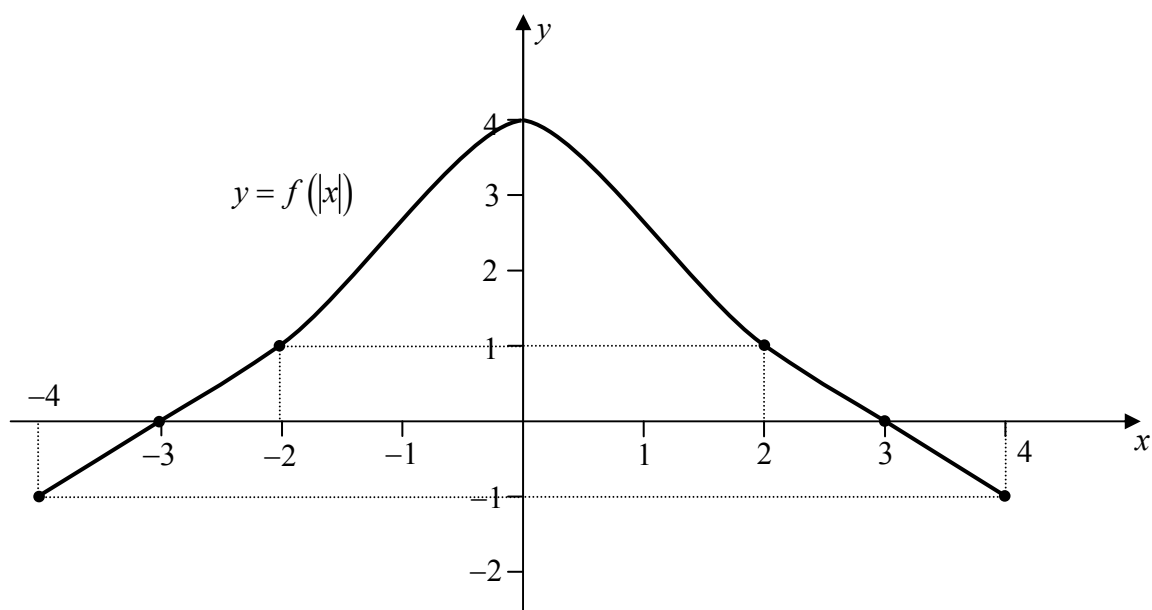
Let $t = \tan \frac{x}{2}$ $x=0, t=0$
 $\tan^{-1} t = \frac{x}{2}$ $x = \frac{\pi}{2}, t=1$
 $x = 2 \tan^{-1} t$
 $dx = \frac{2}{1+t^2} dt$

$$\begin{aligned}
\text{(ii)} \quad \int_0^{2a} f(x) dx &= \int_0^a f(x) dx + \int_a^{2a} f(x) dx && \text{Let } u = 2a - x \quad (\text{in 2}^{\text{nd}} \text{ integral}) \\
& && (\text{so } x = u - 2a) \\
&= \int_0^a f(x) dx + \int_a^0 f(2a - u) \cdot (-du) && du = -dx \\
& && x = 0, \quad u = a \\
& && x = 2a, \quad u = 0 \\
&= \int_0^a f(x) dx + \int_0^a f(2a - u) du \\
&= \int_0^a f(x) dx + \int_0^a f(2a - x) dx \quad (\text{since choice of var does not affect def int}) \\
&= \int_0^a [f(x) + f(2a - x)] dx
\end{aligned}$$

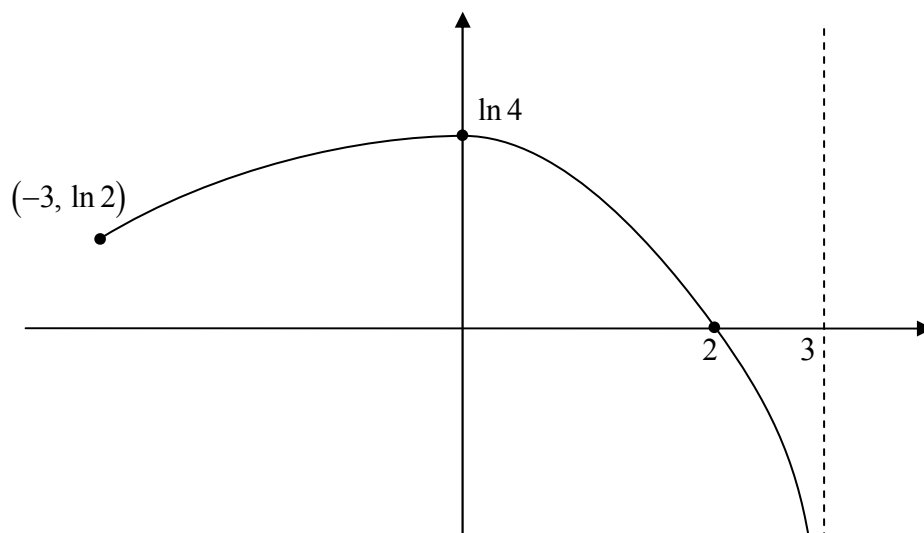
$$\begin{aligned}
\text{(iii)} \quad \int_0^\pi \frac{x}{2 + \sin x} dx &= \int_0^{\frac{\pi}{2}} \left(\frac{x}{2 + \sin x} + \frac{\pi - x}{2 + \sin(\pi - x)} \right) dx && (\text{by part ii}) \\
&= \int_0^{\frac{\pi}{2}} \left(\cancel{\frac{x}{2 + \sin x}} + \frac{\pi}{2 + \sin x} - \cancel{\frac{x}{2 + \sin x}} \right) dx \\
&= \pi \int_0^{\frac{\pi}{2}} \frac{dx}{2 + \sin x} \\
&= \pi \cdot \frac{\pi}{3\sqrt{3}} \quad (\text{by part i}) \\
&= \frac{\pi^2}{3\sqrt{3}}
\end{aligned}$$

Question 14

(a) (i)



(ii)



(b) (i)

$$xy = c^2$$

$$x \cdot \frac{dy}{dx} + y \cdot 1 = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

$$\text{at } P\left(ct, \frac{c}{t}\right), \quad m_T = -\frac{\frac{c}{t}}{ct}$$

$$= -\frac{1}{t^2}$$

Tangent: $y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$

$$t^2 y - ct = -x + ct$$

$$x + t^2 y = 2ct$$

(ii) OT has gradient $\frac{y_1}{x_1}$, and $OT \perp PT$.

$$\text{So } m_{OT} \cdot m_{PT} = -1$$

$$\frac{y_1}{x_1} \cdot \left(-\frac{1}{t^2}\right) = -1$$

$$y_1 = t^2 x_1$$

(iii) Since T satisfies equation of tangent:

$$x_1 + t^2 y_1 = 2ct$$

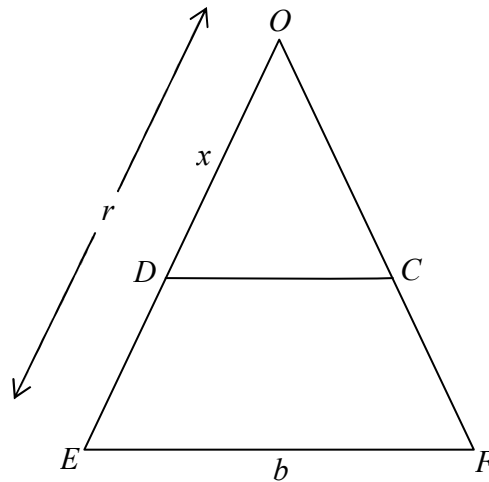
$$x_1 + \frac{y_1}{x_1} \cdot y_1 = 2c \cdot \sqrt{\frac{y_1}{x_1}} \quad (\text{from part ii})$$

$$\begin{aligned} (\times x_1) \quad x_1^2 + y_1^2 &= 2cx_1 \sqrt{\frac{y_1}{x_1}} \\ &= 2c\sqrt{x_1 y_1} \end{aligned}$$

$$(\text{squaring}) \quad (x_1^2 + y_1^2)^2 = 4c^2 x_1 y_1$$

$$\text{ie. locus of } T \text{ is } (x^2 + y^2)^2 = 4c^2 xy$$

(c) (i) By similar triangles OCD and OFE in the base:



$$\frac{DC}{EF} = \frac{OD}{OE}$$

$$\frac{DC}{b} = \frac{x}{r}$$

$$DC = \frac{bx}{r} \quad (\text{base of rectangular slice})$$

Height of slice $h = y$

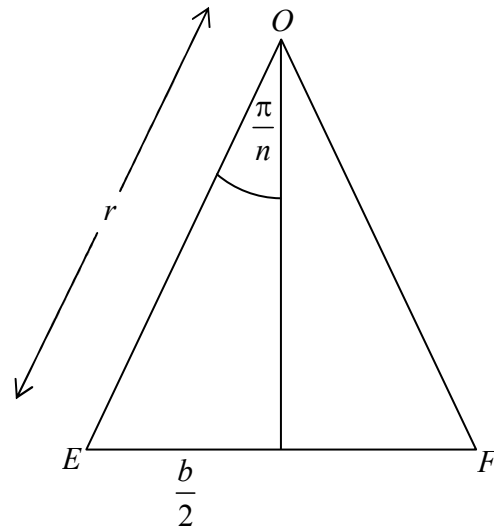
$$= r^2 - x^2$$

Thickness of Slice = δx

$$\therefore \text{ Volume of slice } \delta V = \frac{bx}{r} \cdot (r^2 - x^2) \cdot \delta x$$

$$\begin{aligned}
 \text{(ii) Volume } V &= \lim_{\delta x \rightarrow 0} \sum_{x=0}^r \frac{bx}{r} (r^2 - x^2) \delta x \\
 &= \frac{b}{r} \int_0^r (r^2 x - x^3) dx \\
 &= \frac{b}{4r} [2r^2 x^2 - x^4]_0^r \\
 &= \frac{b}{4r} (2r^4 - r^4) \\
 &= \frac{br^3}{4}
 \end{aligned}$$

(iii)



$$\sin \frac{\pi}{n} = \frac{b}{2r}$$

$$b = 2r \sin \frac{\pi}{n}$$

$$\begin{aligned}
 V_n &= \frac{1}{4} \cdot b \cdot r^3 \\
 &= n \cdot \frac{1}{4} \cdot 2r \sin \frac{\pi}{n} \cdot r^3 \\
 &= \frac{1}{2} r^4 n \sin \frac{\pi}{n}
 \end{aligned}$$

(iv) As $n \rightarrow \infty$, $\frac{\pi}{n} \rightarrow 0$, so $\sin \frac{\pi}{n} \rightarrow \frac{\pi}{n}$

$$\begin{aligned}
 \text{So } \lim_{n \rightarrow \infty} V_n &= \frac{1}{2} r^4 n \cdot \frac{\pi}{n} \\
 &= \frac{1}{2} \pi r^4
 \end{aligned}$$

Question 15

(a) Let $P(x) = 2x^3 - 5x + 1$

$$P\left(-\frac{x}{2}\right) = 0 \text{ has roots } -2\alpha, -2\beta, -2\gamma$$

$$2\left(-\frac{x}{2}\right)^3 - 5\left(-\frac{x}{2}\right) + 1 = 0$$

$$-\frac{x^3}{4} + \frac{5x}{2} + 1 = 0$$

$$x^3 - 10x - 4 = 0$$

(b) (i) $z^n + z^{-n} = (\cos \theta + i \sin \theta)^n + (\cos \theta + i \sin \theta)^{-n}$
 $= \cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta)$
 $= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta$
 $= 2 \cos n\theta$

(ii) $\left(z + \frac{1}{z}\right)^3 = z^3 + 3z + \frac{3}{z} + \frac{1}{z^3}$
 $z^3 + \frac{1}{z^3} = \left(z + \frac{1}{z}\right)^3 - 3\left(z + \frac{1}{z}\right)$
 $= u^3 - 3u$

(iii) IF you HAD to show this result:

$$\left(z + \frac{1}{z}\right)^5 = z^5 + 5z^3 + 10z + \frac{10}{z} + \frac{5}{z^3} + \frac{1}{z^5}$$

$$z^5 + \frac{1}{z^5} = \left(z + \frac{1}{z}\right)^5 - 5\left(z^3 + \frac{1}{z^3}\right) - 10\left(z + \frac{1}{z}\right)$$

$$= u^5 - 5(u^3 - 3u) - 10u$$

$$= u^5 - 5u^3 + 5u$$

$$1 + \cos 10\theta = 1 + (2 \cos^2 5\theta - 1)$$

$$= 2 \cos^2 5\theta$$

$$= \frac{1}{2} (2 \cos 5\theta)^2$$

$$= \frac{1}{2} (z^5 + z^{-5})^2 \quad (\text{from part i})$$

$$= \frac{1}{2} (u^5 - 5u^3 + 5u)^2 \quad (\text{given})$$

$$= \frac{1}{2} [(2 \cos \theta)^5 - 5(2 \cos \theta)^3 + 5(2 \cos \theta)]^2$$

$$= \frac{1}{2} (32 \cos^5 \theta - 40 \cos^3 \theta + 10 \cos \theta)^2$$

$$= 2 (16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta)^2$$

$$(c) \quad (i) \quad \overline{QP} = \overline{QR} \cdot \text{cis} \frac{\pi}{3} \quad \left(\text{since angle in equilateral triangle is } \frac{\pi}{3} \right)$$

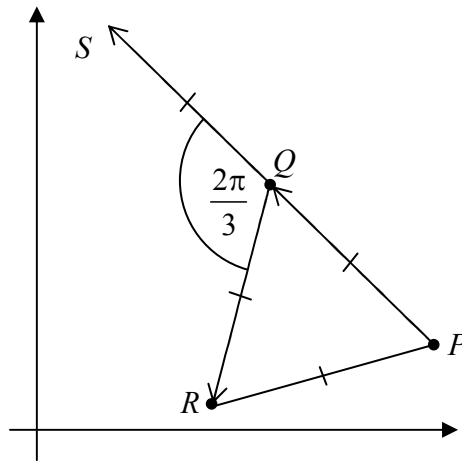
$$p - q = (r - q) \cdot \text{cis} \frac{\pi}{3}$$

$$r - q = (p - q) \cdot \text{cis} \left(-\frac{\pi}{3} \right) \quad \left(\text{to divide by a complex number, multiply by its conjugate} \right)$$

$$r - q = (q - p) \cdot \text{cis} \left(\pi - \frac{\pi}{3} \right) \quad \left(\text{to multiply by } -1, \text{ add } \pi \text{ to the argument} \right)$$

$$r - q = \text{cis} \frac{2\pi}{3} (q - p)$$

OR



$$\angle SQR = \frac{2\pi}{3} \quad \left(\text{exterior angle of triangle} = \text{opposite interior angle} \right)$$

$$r - q = \overline{QR}$$

$$= \text{cis} \frac{2\pi}{3} \cdot \overline{QS} \quad \left(\text{anticlockwise rotation by } \frac{2\pi}{3} \right)$$

$$= \text{cis} \frac{2\pi}{3} \cdot \overline{PQ}$$

$$= \text{cis} \frac{2\pi}{3} (q - p)$$

$$(ii) \quad r - q = \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) (q - p)$$

$$= \left(-\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) (q - p)$$

$$2r - 2q = (-1 + i\sqrt{3})(q - p)$$

$$2r - 2q = -q + p + iq\sqrt{3} - ip\sqrt{3}$$

$$2r = (p + q) + i\sqrt{3}(q - p)$$

- (d) (i) PB subtends equal angles at N and L on the same side of PB .

OR

$$\angle BLP = \angle BNP \text{ (given)}$$

$\therefore BLNP$ is cyclic (converse of angles in same segment [or angles standing on same arc])

(ii)

NOTE: You may NOT say $\angle BNL = \angle MNA$ (vertically opposite)

OR

$$\angle MNP = \angle PBL \text{ (ext angle of cyclic quad = opposite interior angle)}$$

as these assume that LNM is a straight line,

WHICH IS WHAT YOU ARE TRYING TO PROVE.

$$\angle PMA = \angle PNA = 90^\circ \text{ (given)}$$

$\therefore PNAM$ is cyclic (exterior angle of cyclic quad equals opposite interior angle)

$$\angle MNP = \angle MAP \text{ (both angles stand on chord } PM \text{ of cyclic quad } PNAM)$$

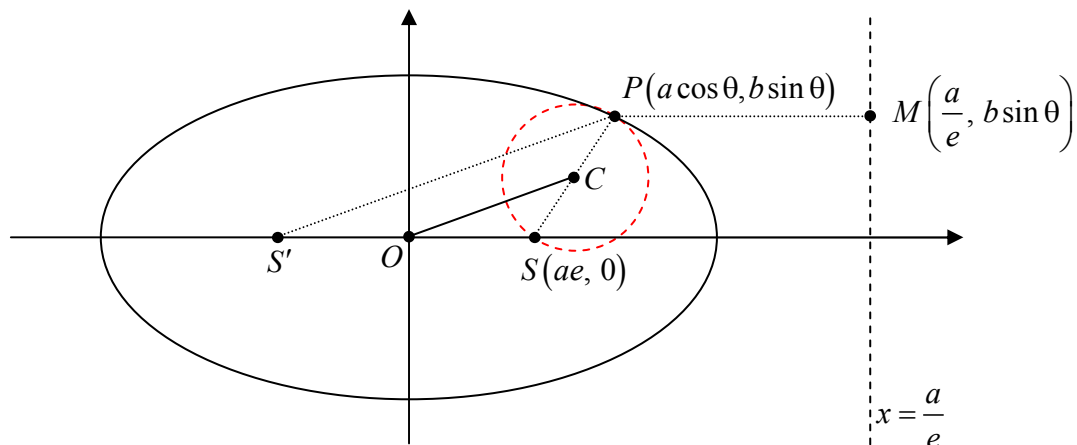
$$= \angle PBC \text{ (exterior angle of cyclic quad } APBC = \text{opposite interior angle)}$$

$\therefore \angle MNP$ is the exterior angle of cyclic quad $BLNP$ (it equals the opposite interior angle)

ie. LNM is straight (ie. L , M and N are collinear)

Question 16

(a) (i)



$$\text{Centre: } C\left(\frac{ae + a \cos \theta}{2}, \frac{b \sin \theta}{2}\right) = C\left(\frac{a(e + \cos \theta)}{2}, \frac{b \sin \theta}{2}\right)$$

$$\text{Diameter: } PS = ePM$$

$$= e\left(\frac{a}{e} - a \cos \theta\right)$$

$$= a(1 - e \cos \theta)$$

$$\therefore \text{radius} = \frac{a}{2}(1 - e \cos \theta)$$

(ii) [The sneaky way]

Since $OS = \frac{1}{2}S'S$, $CS = \frac{1}{2}PS$ and $\angle OSC = \angle S'SC$, then $\triangle OSC$ and $\triangle S'SP$ are similar

$$\therefore OC = \frac{1}{2}PS'$$

But $PS + PS' = 2a$ (sum of focal lengths = length of major axis)

$$\therefore CS + CO = a$$

$$\frac{a}{2}(1 - e \cos \theta) + OC = a$$

$$OC = a - \frac{a}{2} + \frac{a}{2}e \cos \theta$$

$$OC = \frac{a}{2}(1 + e \cos \theta)$$

[The hard slog]

$$OC^2 = \frac{a^2}{4}(e + \cos \theta)^2 + \frac{b^2}{4}\sin^2 \theta$$

$$= \frac{1}{4}(a^2e^2 + 2a^2e \cos \theta + a^2 \cos^2 \theta + b^2 \sin^2 \theta)$$

$$= \frac{1}{4}(a^2e^2 + 2a^2e \cos \theta + a^2 \cos^2 \theta + a^2(1 - e^2)\sin^2 \theta)$$

$$= \frac{a^2}{4}(e^2[1 - \sin^2 \theta] + 2e \cos \theta + [\cos^2 \theta + \sin^2 \theta])$$

$$= \frac{a^2}{4} (e^2 \cos^2 \theta + 2e \cos \theta + 1)$$

$$= \frac{a^2}{4} (e \cos \theta + 1)^2$$

Since $e < 1$ for ellipse

and $|\cos \theta| \leq 1$

then $|e \cos \theta| < 1$

So $1 + e \cos \theta > 0$

$$\therefore OC = \frac{a}{2} (e \cos \theta + 1)$$

$$(b) \quad (i) \quad P\left(-\frac{3}{2}\right) = 4\left(-\frac{3}{2}\right)^3 + 10\left(-\frac{3}{2}\right)^2 + 8\left(-\frac{3}{2}\right) + 3$$

$$= -\frac{27}{2} + \frac{45}{2} - 12 + 3$$

$$= 0$$

$\therefore P(x)$ is divisible by $(2x + 3)$

(ii) Let zeros be $-\frac{3}{2}, \alpha, \beta$

$$\text{Sum:} \quad -\frac{3}{2} + \alpha + \beta = -\frac{5}{2}$$

$$\alpha + \beta = -1$$

$$\text{Product:} \quad -\frac{3}{2} \alpha \beta = -\frac{3}{4}$$

$$\alpha \beta = \frac{1}{2}$$

$\therefore$ A polynomial with zeros α and β is $x^2 + x + \frac{1}{2}$.

But to get equal leading coefficients: $P(x) = (2x + 3)(2x^2 + 2x + 1)$

[Alternatively: divide]

(c) In case you had to prove the given result:

$$\tan(\tan^{-1} x + \tan^{-1} y) = \frac{\tan(\tan^{-1} x) + \tan(\tan^{-1} y)}{1 - \tan(\tan^{-1} x) \cdot \tan(\tan^{-1} y)}$$

$$= \frac{x + y}{1 - xy}$$

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x + y}{1 - xy}$$

The Induction Proof:

$$\text{RTP} \quad \tan^{-1} \frac{1}{2 \times 1^2} + \tan^{-1} \frac{1}{2 \times 2^2} + \tan^{-1} \frac{1}{2 \times 3^2} + \dots + \tan^{-1} \frac{1}{2n^2} = \frac{\pi}{4} - \tan^{-1} \frac{1}{2n+1}$$

$$\text{Test } n=1: \quad \text{LHS} = \tan^{-1} \frac{1}{2}$$

$$\text{RHS} = \frac{\pi}{4} - \tan^{-1} \frac{1}{3}$$

$$\text{LHS} - \text{RHS} = \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} - \frac{\pi}{4}$$

$$= \tan^{-1} \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \cdot \frac{1}{3}} - \frac{\pi}{4} \quad (\text{using given rule})$$

$$= \tan^{-1} 1 - \frac{\pi}{4}$$

$$= \frac{\pi}{4} - \frac{\pi}{4}$$

$$= 0$$

$$\text{LHS} = \text{RHS}$$

$\therefore$ true for $n=1$

Assume true for $n=k$:

$$\text{ie.} \quad \tan^{-1} \frac{1}{2 \times 1^2} + \tan^{-1} \frac{1}{2 \times 2^2} + \tan^{-1} \frac{1}{2 \times 3^2} + \dots + \tan^{-1} \frac{1}{2k^2} = \frac{\pi}{4} - \tan^{-1} \frac{1}{2k+1}$$

Prove true for $n=k+1$:

$$\text{ie. RTP} \quad \tan^{-1} \frac{1}{2 \times 1^2} + \tan^{-1} \frac{1}{2 \times 2^2} + \tan^{-1} \frac{1}{2 \times 3^2} + \dots + \tan^{-1} \frac{1}{2k^2} + \tan^{-1} \frac{1}{2(k+1)^2} = \frac{\pi}{4} - \tan^{-1} \frac{1}{2k+3}$$

$$\text{LHS} - \text{RHS} = \left(\frac{\pi}{4} - \tan^{-1} \frac{1}{2k+1} \right) + \tan^{-1} \frac{1}{2(k+1)^2} - \left(\frac{\pi}{4} - \tan^{-1} \frac{1}{2k+3} \right) \quad (\text{by assumption})$$

$$= \tan^{-1} \frac{1}{2(k+1)^2} + \tan^{-1} \frac{1}{2k+3} - \tan^{-1} \frac{1}{2k+1}$$

$$= \tan^{-1} \frac{\frac{1}{2(k+1)^2} + \frac{1}{2k+3}}{1 - \frac{1}{2(k+1)^2} \cdot \frac{1}{2k+3}} \times \frac{2(k+1)^2(2k+3)}{2(k+1)^2(2k+3)} - \tan^{-1} \frac{1}{2k+1}$$

$$= \tan^{-1} \frac{(2k+3) + 2(k+1)^2}{2(k+1)^2(2k+3) - 1} - \tan^{-1} \frac{1}{2k+1}$$

$$= \tan^{-1} \frac{2k^2 + 6k + 5}{4k^3 + 14k^2 + 16k + 5} - \tan^{-1} \frac{1}{2k+1}$$

$$= \tan^{-1} \frac{2k^2 + 6k + 5}{(2k+1)(2k^2 + 6k + 5)} - \tan^{-1} \frac{1}{2k+1}$$

$$= \tan^{-1} \frac{1}{2k+1} - \tan^{-1} \frac{1}{2k+1}$$

$$= 0$$

$$\text{LHS} = \text{RHS}$$

OR

Using the fact that $\tan^{-1}(-x) = -\tan^{-1} x$:

$$\begin{aligned}
 \text{LHS} &= \frac{\pi}{4} - \tan^{-1} \frac{1}{2k+1} + \tan^{-1} \frac{1}{2(k+1)^2} \\
 &= \frac{\pi}{4} - \left(\tan^{-1} \frac{1}{2k+1} - \tan^{-1} \frac{1}{2(k+1)^2} \right) \\
 &= \frac{\pi}{4} - \tan^{-1} \frac{\frac{1}{2k+1} - \frac{1}{2(k+1)^2}}{1 + \frac{1}{2k+1} \cdot \frac{1}{2(k+1)^2}} \times \frac{2(2k+1)(k+1)^2}{2(2k+1)(k+1)^2} \\
 &= \frac{\pi}{4} - \tan^{-1} \frac{2(k+1)^2 - (2k+1)}{2(2k+1)(k+1)^2 + 1} \\
 &= \frac{\pi}{4} - \tan^{-1} \frac{2k^2 + 2k + 1}{4k^3 + 10k^2 + 8k + 3} \\
 &= \frac{\pi}{4} - \tan^{-1} \frac{1}{2k+3} \quad (\text{by part b}) \\
 &= \text{RHS}
 \end{aligned}$$

$\therefore$ true for $n = k+1$ when true for $n = k$

$\therefore$ by Mathematical Induction, true for all positive integers n .

(d) (i) $f(x) = \log x - x + 1$

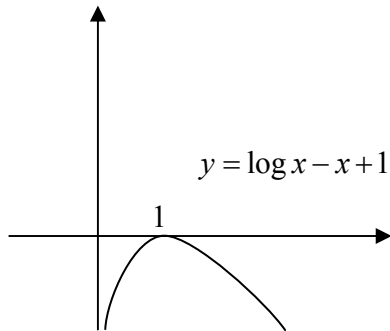
$$\begin{aligned}
 f'(x) &= \frac{1}{x} - 1 \\
 &= \frac{1-x}{x}
 \end{aligned}$$

$\therefore$ stationary point at $x = 1$

$$f''(x) = -\frac{1}{x^2} < 0 \quad \forall x$$

$\therefore$ minimum turning point at $(1, 0)$

Also domain $x > 0$ (and continuous for all x in the domain).



$$\therefore f(x) \leq 0 \quad \forall x > 0$$

(ii) $\sum_{r=1}^n \log(np_r) \leq \sum_{r=1}^n (np_r - 1) \quad (\text{from part i - } \log x \leq x - 1 \quad \forall x > 0)$

$$= \sum_{r=1}^n np_r - \sum_{r=1}^n 1$$

$$\sum_{r=1}^n \log(np_r) \leq \sum_{r=1}^n np_r - n$$

(iii) Continuing from part ii:

$$\sum_{r=1}^n \log np_r \leq n \sum_{r=1}^n p_r - n \quad (\text{since } n \text{ is a constant})$$
$$= n \cdot 1 - n$$

$$\sum_{r=1}^n \log np_r \leq 0$$

(iv) Continuing from part iii:

$$\log np_1 + \log np_2 + \dots + \log np_n \leq 0$$

$$\log(np_1 \cdot np_2 \cdot \dots \cdot np_n) \leq 0$$

$$\log(n^n \cdot p_1 p_2 \dots p_n) \leq 0$$

$$n^n \cdot p_1 p_2 \dots p_n \leq 1$$

Also, since $p_1, p_2, \dots, p_n$ and n are all positive, then $n^n \cdot p_1 p_2 \dots p_n > 0$

$$\therefore 0 < n^n \cdot p_1 p_2 \dots p_n \leq 1$$